

Policies using irrigated area statistics

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Summary

Various institutions, whether implicitly or explicitly, stress the importance of irrigated area datasets for understanding issues such as land degradation, food security, and water scarcity [1–3]. This is evident in initiatives like the FAO’s Sustainable Development Goals [1, 4, 5] and Hand-IN-Hand [6], where irrigated area maps are used to determine areas of degradation or underutilised irrigation, or with the World Bank taking irrigated area data to determine areas that could improve irrigation efficiency and identify sites where farmers can be supported to improve irrigation practices [7]. The link between policy decisions and irrigated area datasets are sometimes less clear; institutions might cite these datasets in their methodology such as the Asian Development Bank [8] when describing how it calculates area equipped for irrigation. Another link between irrigated area datasets and policy decisions are when policy documents reference scientific literature that then incorporates these datasets. Within these examples it is not always evident the extent irrigated area datasets influence decisions made, but a persistent link is still present.

A commonality between the policy documents reviewed is the use of only 1 irrigated area dataset. Rarely it seems will multiple datasets be considered and often the same dataset is cited, that being GMIA likely as it is the dataset put forward by the FAO. Considering the disagreement between different irrigated area datasets demonstrated in this paper, reliance on only one dataset implies the uncertainty of irrigated area is not considered and that decisions are being made with a narrowed particular view of reality.

FAO

1. Sustainable Development Goals

- FAO tracks progress toward several Sustainable Development Goals (SDGs). Using the Global Agro-Ecological Zones (GAEZv5) irrigated-area map, together with land-cover and land-degradation layers, FAO monitor the share of agricultural area under productive and sustainable agriculture (SDG 2.4.1) and the proportion of land that is degraded (SDG15.3.1). These indicators let FAO assess country performance, highlight areas at risk of human-induced land degradation and support actions to restore degraded land and improve food-security outcomes [1, 4, 5].

2. FAO Hand-In-Hand

- The FAO Hand-in-Hand (HIH) initiative [9] provides support in channeling investments to the world’s most vulnerable regions. HIH offers an open-access database that combines geospatial, biophysical and socio-economic data, enabling governments, donors and private-sector partners to locate territories where agricultural investment can most effectively reduce poverty and hunger. The platform uses GAEZv5 to identify regions where irrigation potential is high but under-utilised. The platform has already been used in countries such as Ethiopia, Guatemala, and Southern Africa to guide investment decisions and accelerate rural development [6].

World Bank

1. Prospects for Irrigated Agriculture Whether Irrigated Area and Irrigation Water Must Increase to Meet Food Needs in the Future [10]
 - This report derives from the importance of water for irrigation to the question, how will additional food be produced as competition for scarce land and water resources increases? The International Water Management Institute (IWMI), Food and Agriculture Organization of the United Nations (FAO), and International Food Policy Research Institute (IFPRI) provide a partial picture of supply and demand for food supply and irrigation-water by 2025 and 2030. The World Bank initiated a validation exercise in cooperation with these agencies to ensure that the models: consider a balanced range of assumptions and scenarios; introduce additional assumptions and scenarios; and replace, or fine-tune, some of them. It attempts to introduce into the forecasts the possible impacts of additional policy interventions and to evaluate their likely effects on the global projections of agriculture-water supply and demand. The validation exercise provides a review of the existing model structure and components, the assumptions made, the scenarios, and scenario results. Model assumptions, scenarios, and results of the revised response from the IWMI, FAO, and IFPRI are given as a part of the validation exercise as well as assumptions made and scenarios developed under different criteria and issues as raised by the group of experts.

2. Shaping the Future of Water for Agriculture : A Sourcebook for Investment in Agricultural Water Management. Agriculture and Rural Development [7]

- Agricultural water management is a vital practice in ensuring reduction, and environmental protection. After decades of successfully expanding irrigation and improving productivity, farmers and managers face an emerging crisis in the form of poorly performing irrigation schemes, slow modernization, declining investment, constrained water availability, and environmental degradation. More and better investments in agricultural water are needed. In response, the World Bank, in conjunction with many partner agencies, has compiled a selection of good experiences that can guide practitioners in the design of quality investments in agricultural water. The messages of this publication center around the key challenges to agricultural water management, specifically: building policies and incentives; designing institutional reforms; investing in irrigation systems improvement and modernization; investing in groundwater irrigation; investing in drainage and water quality management; investing in water management in rain-fed agriculture; investing in agricultural water management in multipurpose operations; coping with extreme climatic conditions; and assessing the social, economic, and environmental impacts of agricultural water investments.
- "The amount of water that could be saved by 2025 by achieving 70 percent irrigation efficiency on the world's gross irrigated area could meet about half the demand for additional water supplies (FAO 2001b). Such savings are the main option for addressing water shortage challenges in many developing countries", page 105. NOTE: Cannot find exactly where they source irrigated area data from but they are using it to define this above metric. I.e. improve irrigation efficiency in these defined irrigated areas to help water shortages in the majority world.

3. Uzbekistan. Policy Perspectives for Irrigation and Drainage Sector Reform [11]

- This policy note explores the potential for a service-oriented, performance-based approach to accelerate the transformation of the I&D sector. It is organized into six sections. Section 1 identifies five factors of Uzbekistan's country context that determine the urgent need for reform. Section 2 reviews sector characteristics and describes the benefits to be reaped from transformation of the sector. Section 3 focuses on investment trends and gaps. Section 4 describes a service-oriented, performance-based approach to I&D reform. Section 5 describes enablers of reform. Section 6 identifies a set of next steps for moving this agenda forward, including a framework for development partners to engage in this process.
- Sources irrigated area data from Oblast, 2019 and FAO AQUASTAT, made evident by Figure 4 on page 7

4. The Farmer-led Irrigation Development Guide : A What, Why and How-to for Intervention Design (English) [12]

- Water is essential for communities for drinking, for growing the food we eat and sell, for supporting the ecosystems within which we thrive. Ensuring sustainable management and access to water and sanitation for all is key for resilient livelihoods and for achieving the Sustainable Development Goals. Climate shocks, inequitable access to water resources, critical weaknesses in the resilience of the natural resources on which much of human livelihood depends and economic woes amplified by the Coronavirus (COVID-19) pandemic have resulted increasingly in institutional and economic water scarcity. This impacts heavily on farmers, particularly in Africa, disrupting their capacity to produce and sell and severely affecting income and livelihoods. Scaling up and accelerating Farmer-led Irrigation Development (FLID) holds significant potential for short-term recovery for the most vulnerable farmers and strengthen farmers' ability to recover from shocks and adapt to a changing environment. FLID recognizes that farmers have persistently filled the gap where public-led irrigation services haven't reached and improved on them where they do exist. This guide aims to provide governments with the practical strategies to catalyze FLID and scale it up, making it a faster.
- "Remote sensing and satellite data analytics of irrigated areas and water resource use is being used to identify where and how to support farmers to improve irrigated practices", page M5-32. References other studies but not a particular dataset(s), however understanding of irrigated area is important to their perceived "benefits of managing soil water are substantial in water, productivity and financial terms [...]" Expanded irrigated areas with the same equipment and labor due to overall lower irrigation applications" page M7-27-28. Figure 2 on page 58 also plots current irrigated area and trends and targets of future irrigated area.

Consultative Group on International Agricultural Research

1. Watering the future: The economics of sustainable irrigation [13]
 - Discusses the economic viability of sustainable irrigation in smallholder agriculture
 - The research article cited for this is 'Economics of sustainable irrigation in smallholder agriculture: implications for food security and climate action' [14]
 - The above article's evidence base includes 'Half of twenty-first century global irrigation expansion has been in water-stressed regions'. The data for area equipped for irrigation used within this article comes from FAO-STAT and AQUASTAT (both derived from GMIA) as well as EURO-STAT (taken from survey data) [15]
 - This linkage is an example of how irrigated area databases are picked up by scientific literature and eventually tie back into policy decisions

Asian Development Bank

1. ASIAN WATER DEVELOPMENT OUTLOOK 2020: Description of Methodology and Data [8]
 - This document outlines the methodology and data collection used by the Asian Development Bank
 - In citing FAOSTAT 2016 for land equipped for irrigation, the document makes clear that the FAO’s irrigated area dataset (GMIA) plays a role in decision making

International Fund for Agricultural Development

Both IFAD and AIIB (Asian Infrastructure Investment Bank) co-financed the initial project of CAISAR (Climate Adaptive Irrigation and Sustainable Agriculture for Resilience) in Cambodia aimed at improving climate resilience in rural areas of the country. In order to assess water management initiatives, the Ministry of Water Resources and Meteorology of Cambodia (MOWRAM) is establishing the Water Resources Data Management Center (NWRDMC) and a National Water Resources Information System (WRIS) with the assistance of the Asian Development Bank.

WRIS integrates **spatial information**, CISIS and IAMS, meteorological data, hydrological data, ground measurements, **satellite-derived remote sensing data**, and Lidar. All data generated on the CAISAR, and all watersheds, are coalesced in WRIS and used to model water availability, use, and floods across entire watersheds.”

- "WRIS(...) uses satellite-derived remote sensing data, Lidar, (...) satellite-based information, global and regional data sets and results from the modelling systems in other projects"[16]

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